

International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Mid Term Examination — Solutions

Total Marks: 40

Duration: 90 Minutes

Disclaimer

1. The solutions given here are not the only ones possible.
2. Results from the book used below are referred to by their numbers in Tao, *Analysis I*: Lemma 3.1.6 (single choice), Proposition 4.3.3 (properties of absolute value and distance) and Proposition 4.4.1 (integer part, which gives the Archimedean property).

Question (1) [10 Marks]. You may use the following fact without proof:

Let $p = (x, y)$ be an ordered pair. Then for all sets X and Y , $p \in X \times Y \iff (x \in X \text{ and } y \in Y)$.

Let A, B, C, D be non-empty sets. Then

- (a) [5 Marks] Show that $A \times B \subseteq C \times D$ if and only if $A \subseteq C$ and $B \subseteq D$.
- (b) [5 Marks] Show that $A \times B = C \times D$ if and only if $A = C$ and $B = D$.

Solution. Throughout we use the fact stated in the question, which we call (*): for an ordered pair (x, y) and sets X, Y , $(x, y) \in X \times Y \iff (x \in X \text{ and } y \in Y)$. Recall also that, by definition of the Cartesian product, every element of $X \times Y$ is an ordered pair (x, y) with $x \in X$ and $y \in Y$.

(a) ($\Leftarrow$) Suppose $A \subseteq C$ and $B \subseteq D$, and let $p \in A \times B$. Then $p = (x, y)$ with $x \in A$ and $y \in B$. Hence $x \in C$ and $y \in D$, so $p = (x, y) \in C \times D$ by (*). Thus $A \times B \subseteq C \times D$. (This direction does not use non-emptiness.)

($\Rightarrow$) Suppose $A \times B \subseteq C \times D$.

$A \subseteq C$. Let $x \in A$. Since B is non-empty, by the single choice lemma (Lemma 3.1.6) we can choose some $y \in B$. By (*), $(x, y) \in A \times B$, hence $(x, y) \in C \times D$, and by (*) again $x \in C$. As $x \in A$ was arbitrary, $A \subseteq C$.

$B \subseteq D$. Let $y \in B$. Since A is non-empty, choose some $x \in A$. Then $(x, y) \in A \times B \subseteq C \times D$, so $y \in D$ by (*). Hence $B \subseteq D$.

(b) ($\Leftarrow$) If $A = C$ and $B = D$, then $A \times B$ and $C \times D$ are the same set.

($\Rightarrow$) Suppose $A \times B = C \times D$. Then $A \times B \subseteq C \times D$, and since A, B are non-empty, part (a) gives $A \subseteq C$ and $B \subseteq D$. Also $C \times D \subseteq A \times B$, and since C, D are non-empty, part (a) with the roles of the two products exchanged gives $C \subseteq A$ and $D \subseteq B$. Hence $A = C$ and $B = D$, since two sets that are each contained in the other are equal.

Remark. Only $A \neq \emptyset$ and $B \neq \emptyset$ are really needed in (b): if $x \in A$ and $y \in B$, then $(x, y) \in A \times B = C \times D$, so C and D are automatically non-empty. Without the non-emptiness hypothesis both statements fail; for instance $\emptyset \times \{0\} = \emptyset = \emptyset \times \{1\}$ although $\{0\} \neq \{1\}$.

Question (2) [10 Marks]. Define two sequences of natural numbers by $p_1 := 1$, $q_1 := 1$ and

$$p_{n+1} := p_n + 2q_n, \quad q_{n+1} := p_n + q_n.$$

- (a) [5 Marks] Prove that $p_n \geq q_n \geq 1$ for every $n \geq 1$, and that $p_n^2 - 2q_n^2 = (-1)^n$ (an identity between integers).
- (b) [5 Marks] Deduce that $q_{n+1} \geq 2q_n$, hence $q_n \geq 2^{n-1}$, hence $q_n \geq n$ for every $n \geq 1$.

Solution. The first few values are $(p_n, q_n) = (1, 1), (3, 2), (7, 5), (17, 12), (41, 29), \dots$, which is a useful check on everything below.

(a) *The inequalities, by induction on n .* For $n = 1$ we have $p_1 = q_1 = 1$, so $p_1 \geq q_1 \geq 1$.

Suppose $p_n \geq q_n \geq 1$ for some $n \geq 1$. Then

$$q_{n+1} = p_n + q_n \geq q_n \geq 1,$$

since adding the natural number p_n cannot decrease q_n . Next,

$$p_{n+1} = p_n + 2q_n = (p_n + q_n) + q_n = q_{n+1} + q_n \geq q_{n+1}.$$

So $p_{n+1} \geq q_{n+1} \geq 1$, which closes the induction.

The identity, by induction on n . The identity is a statement in $\mathbb{Z}$, since its right-hand side may be -1 . For $n = 1$, $p_1^2 - 2q_1^2 = 1 - 2 = -1 = (-1)^1$.

Suppose $p_n^2 - 2q_n^2 = (-1)^n$ for some $n \geq 1$. Expanding,

$$\begin{aligned} p_{n+1}^2 - 2q_{n+1}^2 &= (p_n + 2q_n)^2 - 2(p_n + q_n)^2 \\ &= (p_n^2 + 4p_nq_n + 4q_n^2) - (2p_n^2 + 4p_nq_n + 2q_n^2) \\ &= -p_n^2 + 2q_n^2 = -(p_n^2 - 2q_n^2) = -(-1)^n = (-1)^{n+1}, \end{aligned}$$

using the induction hypothesis in the second-to-last step. This closes the induction.

(b) $q_{n+1} \geq 2q_n$. By (a), $p_n \geq q_n$, and adding q_n to both sides preserves the order, so

$$q_{n+1} = p_n + q_n \geq q_n + q_n = 2q_n.$$

$q_n \geq 2^{n-1}$, *by induction*. For $n = 1$: $q_1 = 1 = 2^0$. If $q_n \geq 2^{n-1}$, then multiplying by 2 preserves the order, so $q_{n+1} \geq 2q_n \geq 2 \cdot 2^{n-1} = 2^n$.

$2^{n-1} \geq n$, *by induction*. For $n = 1$: $2^0 = 1 \geq 1$. If $2^{n-1} \geq n$ for some $n \geq 1$, then

$$2^n = 2 \cdot 2^{n-1} \geq 2n = n + n \geq n + 1,$$

the last step because $n \geq 1$.

Combining the last two facts, $q_n \geq 2^{n-1} \geq n$ for every $n \geq 1$.

Remark. Writing $x_n := p_n/q_n$ and dividing the identity in (a) by q_n^2 gives $x_n^2 - 2 = (-1)^n/q_n^2$, so by (b), $|x_n^2 - 2| = 1/q_n^2 \leq 1/n^2$. The fractions $x_n = 1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \dots$ therefore have squares as close to 2 as we like, although by Proposition 4.4.4 no rational has square exactly 2.

Question (3) [10 Marks]. Let $(a_n)_{n \geq 1}$ be a sequence of rationals. For a given rational $\varepsilon > 0$, the sequence $(a_n)_{n \geq 1}$ is said to be eventually ε -steady iff there exists a natural number $N \geq 1$ such that $|a_i - a_j| \leq \varepsilon$ for all $i, j \geq N$.

- (a) [5 Marks] For every rational $\varepsilon > 0$, give an example of an eventually ε -steady sequence of rationals which is not a Cauchy sequence. Justify your answer.
- (b) [5 Marks] Is the sequence $(a_n)_{n \geq 1}$, where $a_n = (-1)^n$ for all $n \geq 1$, a Cauchy sequence? Justify your answer.

Solution. Recall that a sequence is Cauchy if and only if it is eventually δ -steady for *every* rational $\delta > 0$. So to show a sequence is not Cauchy, it suffices to exhibit one rational $\delta > 0$ for which it is not eventually δ -steady.

(a) Let $\varepsilon > 0$ be an arbitrary rational, and define

$$a_n := (-1)^n \frac{\varepsilon}{2} \quad (n \geq 1),$$

so the sequence is $-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}, -\frac{\varepsilon}{2}, \frac{\varepsilon}{2}, \dots$

Eventually ε -steady. Let $i, j \geq 1$. If i and j have the same parity then $a_i = a_j$ and $|a_i - a_j| = 0 \leq \varepsilon$. If they have different parity then one of a_i, a_j is $\varepsilon/2$ and the other is $-\varepsilon/2$, so $|a_i - a_j| = \varepsilon \leq \varepsilon$. So $|a_i - a_j| \leq \varepsilon$ for all $i, j \geq 1$, and $N := 1$ works.

Not Cauchy. Take $\delta := \varepsilon/2$, a positive rational. Let $N \geq 1$ be arbitrary. The indices $i := N$ and $j := N + 1$ are both at least N and have different parity, so $|a_i - a_j| = \varepsilon > \varepsilon/2 = \delta$. Hence no N works for δ , the sequence is not eventually δ -steady, and so it is not Cauchy.

(b) No. Take $\delta := 1$ and let $N \geq 1$ be arbitrary. With $i := N$ and $j := N + 1$, both at least N ,

$$|a_i - a_j| = |(-1)^N - (-1)^{N+1}| = |(-1)^N| \cdot |1 - (-1)| = 2 > 1.$$

So the sequence is not eventually 1-steady, and hence it is not Cauchy.

Remark. The example in (a) must depend on ε . If a single sequence were eventually ε -steady for every rational $\varepsilon > 0$, it would be Cauchy by definition. For instance $(-1)^n$ from part (b) is eventually ε -steady exactly when $\varepsilon \geq 2$, so it answers (a) only for $\varepsilon \geq 2$. An answer that uses $(-1)^n$ and justifies it only for such ε is incomplete.

Question (4) [10 Marks]. A sequence $(a_n)_{n \geq 1}$ of rationals is called *anchored* if for every rational $\varepsilon > 0$ there exists natural $N \geq 1$ with

$$|a_j - a_N| \leq \varepsilon \quad \text{for all } j \geq N.$$

Prove that a sequence of rationals is anchored if and only if it is a Cauchy sequence.

Solution. ($\Leftarrow$) *Cauchy implies anchored.* Let (a_n) be Cauchy and let $\varepsilon > 0$ be rational. By definition there is $N \geq 1$ with $|a_i - a_j| \leq \varepsilon$ for all $i, j \geq N$. Taking $i := N$, which satisfies $N \geq N$, gives $|a_j - a_N| \leq \varepsilon$ for all $j \geq N$. So (a_n) is anchored.

($\Rightarrow$) *Anchored implies Cauchy.* Let (a_n) be anchored and let $\varepsilon > 0$ be rational. Then $\varepsilon/2$ is also a positive rational, so by the anchored property there is $N \geq 1$ with

$$|a_j - a_N| \leq \frac{\varepsilon}{2} \quad \text{for all } j \geq N.$$

Now let $i, j \geq N$. By the triangle inequality (Proposition 4.3.3),

$$|a_i - a_j| = |(a_i - a_N) + (a_N - a_j)| \leq |a_i - a_N| + |a_j - a_N| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

where we used $|a_N - a_j| = |a_j - a_N|$. So (a_n) is eventually ε -steady. As ε was an arbitrary positive rational, (a_n) is Cauchy.

Remark. The anchored property must be applied with $\varepsilon/2$, not ε . Applying it with ε only gives $|a_i - a_j| \leq 2\varepsilon$, which does not match the definition of a Cauchy sequence without a further argument.

Question (5) [10 Marks]. A function $D : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ is said to be a distance measure on $\mathbb{Q}$ if for all $a, b, c \in \mathbb{Q}$: (1) $D(a, b) \geq 0$ and equality holds iff $a = b$. (2) $D(a, b) = D(b, a)$. (3) $D(a, b) \leq D(a, c) + D(c, b)$. A sequence $(a_n)_{n \geq 1}$ of rationals is called D -Cauchy iff

$$\forall \text{ rational } \varepsilon > 0 \quad \exists \text{ natural } N \geq 1 \text{ such that } D(a_i, a_j) \leq \varepsilon \quad \forall i, j \geq N.$$

Then for $D(a, b) = |a - b|$, we get the usual definition of Cauchy sequences. Now define a function $\phi : \mathbb{Q} \rightarrow \mathbb{Q}$ by $\phi(x) = 1/x$ for all nonzero rationals x and $\phi(0) = 0$. Then define $D_\phi(a, b) = |\phi(a) - \phi(b)|$ for all $a, b \in \mathbb{Q}$.

- (a) [5 Marks] Prove that D_ϕ is a valid distance measure on $\mathbb{Q}$.
- (b) [5 Marks] Give an example of a sequence of rationals that is D_ϕ -Cauchy but not a Cauchy sequence. Justify your answer.

Solution.

(a) First, ϕ is a function from $\mathbb{Q}$ to $\mathbb{Q}$: for $x \neq 0$ the reciprocal $1/x$ is a rational number, and $\phi(0) = 0$. Hence $D_\phi(a, b)$ is a rational number for all $a, b \in \mathbb{Q}$.

ϕ is injective. Suppose $\phi(a) = \phi(b)$. We first note that $\phi(x) \neq 0$ whenever $x \neq 0$: if $1/x = 0$ then $1 = x \cdot (1/x) = x \cdot 0 = 0$, which is false. Now consider cases.

- If $a = 0$ and $b = 0$, then $a = b$.
- If exactly one of a, b is 0, say $a = 0 \neq b$, then $\phi(a) = 0 \neq \phi(b)$, contradicting $\phi(a) = \phi(b)$. So this case cannot occur.
- If $a \neq 0$ and $b \neq 0$, then $1/a = 1/b$; multiplying both sides by ab gives $b = a$.

In every possible case $a = b$.

We now verify the three properties, writing $x := \phi(a)$, $y := \phi(b)$, $z := \phi(c)$.

(1) $D_\phi(a, b) = |x - y| \geq 0$ by Proposition 4.3.3(a). Moreover $|x - y| = 0$ if and only if $x = y$, again by Proposition 4.3.3(a). Now $x = y$ means $\phi(a) = \phi(b)$, which holds if and only if $a = b$: the “if” direction is clear, and the “only if” direction is injectivity of ϕ . So $D_\phi(a, b) = 0$ if and only if $a = b$.

(2) $D_\phi(a, b) = |x - y| = |-(y - x)| = |y - x| = D_\phi(b, a)$, using $|-w| = |w|$ from Proposition 4.3.3(d).

(3) By the triangle inequality for distance, Proposition 4.3.3(g),

$$D_\phi(a, b) = |x - y| \leq |x - z| + |z - y| = D_\phi(a, c) + D_\phi(c, b).$$

So D_ϕ is a distance measure on $\mathbb{Q}$.

(b) Let $a_n := n$ for $n \geq 1$. Since $n \neq 0$, $\phi(a_n) = 1/n$, so

$$D_\phi(a_i, a_j) = \left| \frac{1}{i} - \frac{1}{j} \right|.$$

(a_n) is D_ϕ -Cauchy. Let $\varepsilon > 0$ be rational. By the Archimedean property (Proposition 4.4.1) there is an integer M with $M \leq 1/\varepsilon < M + 1$. Since $1/\varepsilon > 0$ we have $M + 1 > 0$, so $N := M + 1$ is a natural number with $N \geq 1$ and $N > 1/\varepsilon$. Multiplying by the positive rational ε/N gives $1/N < \varepsilon$.

Now let $i, j \geq N$. By symmetry we may assume $i \leq j$. Then $0 < 1/j \leq 1/i \leq 1/N$, so

$$D_\phi(a_i, a_j) = \frac{1}{i} - \frac{1}{j} \leq \frac{1}{i} \leq \frac{1}{N} < \varepsilon.$$

Hence (a_n) is D_ϕ -Cauchy. (Equivalently: $(1/n)$ is a Cauchy sequence, Proposition 5.1.11 of Tao, and $D_\phi(a_i, a_j)$ is exactly $|1/i - 1/j|$.)

(a_n) is not Cauchy. Take $\delta := 1/2$ and let $N \geq 1$ be arbitrary. With $i := N + 1$ and $j := N$, both at least N ,

$$|a_i - a_j| = |(N + 1) - N| = 1 > \frac{1}{2}.$$

So (a_n) is not eventually $\frac{1}{2}$ -steady, and hence not Cauchy.

Remark. Two further observations. First, $D_\phi(n, 0) = |1/n - 0| = 1/n$, so measured with D_ϕ the sequence $1, 2, 3, \dots$ approaches 0. Second, the implication also fails the other way for the same distance: $(1/n)$ is Cauchy, but $D_\phi(1/i, 1/j) = |i - j|$, so it is not D_ϕ -Cauchy. Whether a sequence is Cauchy depends on how distance is measured, not only on the sequence itself; the real numbers are built from the particular choice $D(a, b) = |a - b|$.